

# INSTALLATION INSTRUCTION

## PAL LIGHTING PCR-2DMX LIGHTING CONTROLLER



64-PCR-2DMX



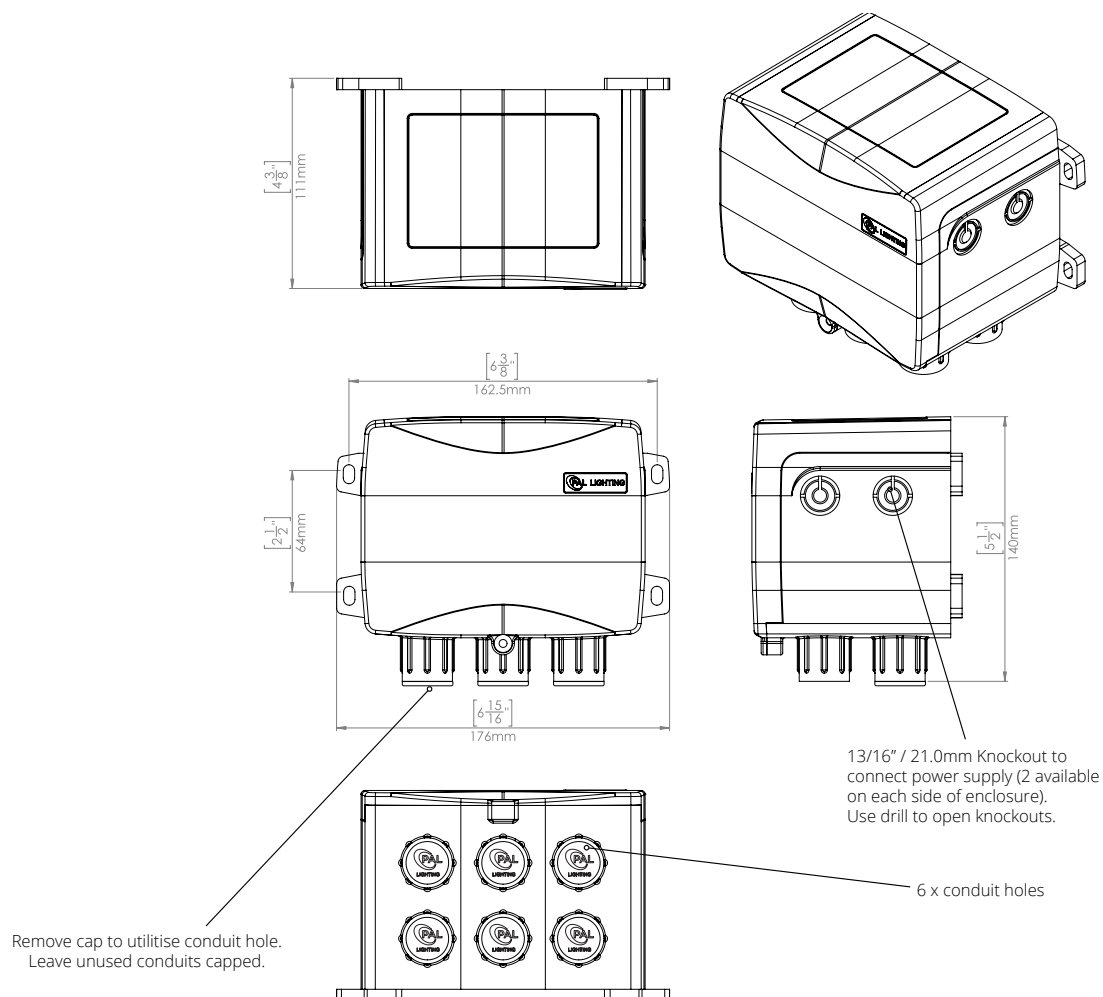
### IMPORTANT INFORMATION

#### WARNING

- This PCR-2DMX Class 2 power supply is specifically designed to supply 24V DC to power the PAL 4 Wire UL Listed LED Pool lights.
- The PCR-2DMX must be installed in accordance with the National Electrical Code by a Certified Electrician or Qualified Pool Technician.
- PAL drivers are DMX receivers, not controllers. A DMX controller is required between any home automation system and a PAL DMX driver. Connection Hierarchy: Home Automation System → DMX Controller → PAL DMX Driver

NOTE: No bonding is required from this Class 2 power supply to the UL Listed PAL Pool Light fixtures as they are made of an all plastic construction and comply with NEC article 680 requirements.

### DIMENSIONS



#### Knockout Instructions

It is recommended that the cover is fitted and the cover screws tight when the knockouts are being removed.

When removing the knock outs place a flat blade screwdriver with a 3/16" (4mm) blade as shown and a light hit with a hammer to the head of the screw driver will allow the Knockout to be removed.

Alternatively drill location points are located in the centre of each Knock out to allow the use of a hole saw to remove the knockouts.

## INSTALLATION INSTRUCTIONS FOR 16/35/65 WATT PCR-2DMX DRIVER/RECEIVER

**Step 1.** The PCR-2DMX Class 2 Power Supply is a Weatherproof construction (IP65) and should be mounted on a flat surface using the mounting legs on the enclosure with appropriate fixtures for the mounting substrate and in a suitable position to suit the intended location of the PAL 4 Wire LED Pool Lights and installed to comply with all the requirements of the National Electrical Code (NEC) and all relevant local codes and ordinances.

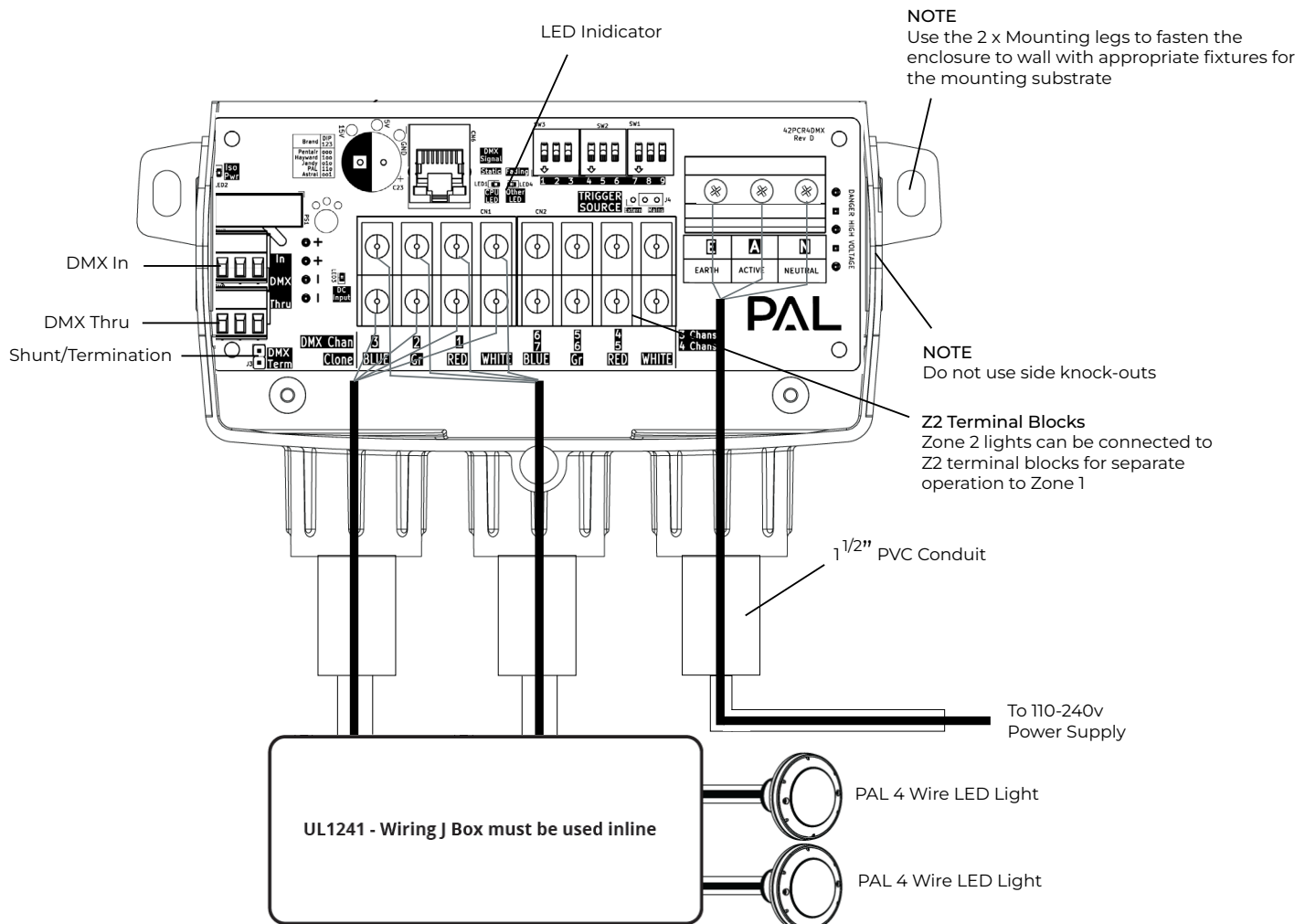
NOTE: The Driver Enclosure must be located a minimum distance of 5 feet measured horizontally from the inside wall of the pool and not less than 1 foot above the maximum pool water level measured to the top of the Driver Enclosure.

NOTE: The enclosure must be mounted using the mounting legs on the enclosure with the screw set supplied. If the enclosure is to be mounted on a concrete or block wall, mark out the mounting hole locations and drill 3/16" holes using a masonry drill then fit the Star Plugs supplied into the holes and fix using the stainless steel screws supplied.

**Step 2.** Using the conduit entry on the bottom (bottom-front on PCR-2DMX) right hand side of the PCR-2DMX (as shown) connect the 110-240V power supply to NEC code requirements.

**Step 3.** Using the remaining conduit entries at the bottom of the PCR-2DMX (as shown), connect the 4 core wire from the PAL 4 WIRE LED LIGHTS. Please ensure the 4 cores are connected to the correct terminals. 2 x output conduit entries are provided in the enclosure to connect up to 2 x PAL-LED 4 Wire Pool Lights.

### V3 16/35/65 WATT PCR-2DMX DRIVER/RECEIVER



DMX Connection

Connect the DMX output wires into the lighting controller using the “In” terminals.

If connecting to an additional lighting controller or device in the same DMX universe, use the “Thru” terminals to run the wires to the next device(s).

If NO additional devices are being added, the jumper bridge/shunt MUST be on the “DMX Term”/J3 pins.



A “closed” universe.  
Prevents loss of  
DMX signal.



No jumper to allow  
DMX signal to go  
to the next connect-  
ed

“Long Strip” Mode

Dipswitch #9 toggles “long strip” mode. This is used when an LED strip is used for long lengths. Pool perimeter, long hardscape runs, etc.



| Mode           | #9  | Note                                                  |
|----------------|-----|-------------------------------------------------------|
| Long Strip Off | Off | Normal                                                |
| Long Strip On  | ON  | The output is altered to better suit long RGB strips. |

Long strip mode will lower overall brightness to better suit longer strip runs. The length is strip dependant although 30+ feet is a good general guideline.

3 Channel or 4 Channel DMX Addressability

Dip switch #7 is designed to toggle between 3 channel and 4 channel addressability for our lights in a zone. This is toggled to help accommodate different addresses needed within a DMX universe and to suit DMX controllers which only have 4 channel light libraries.

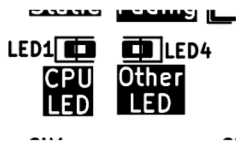
- 3 channel allows for addresses 1 - 379 (Dipswitch #7 is off)
- 4 channel allows for addresses 1 - 505 (Dipswitch #7 is on)

Refer to the tables on the following pages for dipswitch configurations.



Software Information

The CPU LED will typically show red color. This will also provide software information at bootup/restart.



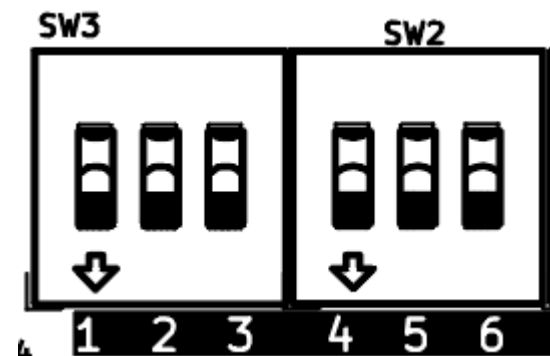
CPU LED uses short pulses to indicate numbers. (I.E. 2 short pulses indicates the number 2.)

Other LED uses a long pulse to indicate a decimal.

Example:  
2 short pulses on the CPU LED followed by 1 long pulse on the Other LED followed by 1 short pulse on the CPU LED indicates 2.1

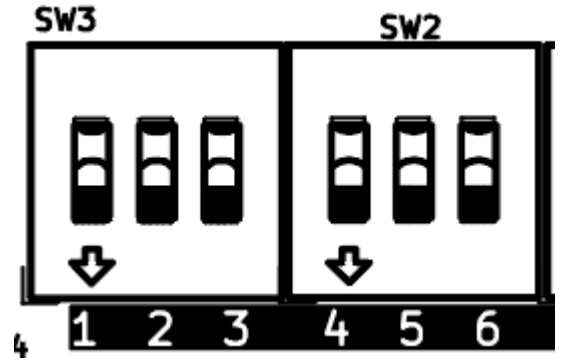
### V3 PCR-2DMX DRIVER/RECEIVER - 3 Channel addresses (Dipswitch #7 is off)

| Start Address | SW3 #1 "1" | SW3 #2 "2" | SW3 #3 "3" | SW2 #1 "4" | SW2 #2 "5" | SW2 #3 "6" |
|---------------|------------|------------|------------|------------|------------|------------|
| 1             | Off        | Off        | Off        | Off        | Off        | Off        |
| 7             | Off        | Off        | Off        | Off        | Off        | On         |
| 13            | Off        | Off        | Off        | Off        | On         | Off        |
| 19            | Off        | Off        | Off        | Off        | On         | On         |
| 25            | Off        | Off        | Off        | On         | Off        | Off        |
| 31            | Off        | Off        | Off        | On         | Off        | On         |
| 37            | Off        | Off        | Off        | On         | On         | Off        |
| 43            | Off        | Off        | Off        | On         | On         | On         |
| 49            | Off        | Off        | On         | Off        | Off        | Off        |
| 55            | Off        | Off        | On         | Off        | Off        | On         |
| 61            | Off        | Off        | On         | Off        | On         | Off        |
| 67            | Off        | Off        | On         | Off        | On         | On         |
| 73            | Off        | Off        | On         | On         | Off        | Off        |
| 79            | Off        | Off        | On         | On         | Off        | On         |
| 85            | Off        | Off        | On         | On         | On         | Off        |
| 91            | Off        | Off        | On         | On         | On         | On         |
| 97            | Off        | On         | Off        | Off        | Off        | Off        |
| 103           | Off        | On         | Off        | Off        | Off        | On         |
| 109           | Off        | On         | Off        | Off        | On         | Off        |
| 115           | Off        | On         | Off        | Off        | On         | On         |
| 121           | Off        | On         | Off        | On         | Off        | Off        |
| 127           | Off        | On         | Off        | On         | Off        | On         |
| 133           | Off        | On         | Off        | On         | On         | Off        |
| 139           | Off        | On         | Off        | On         | On         | On         |
| 145           | Off        | On         | On         | Off        | Off        | Off        |
| 151           | Off        | On         | On         | Off        | Off        | On         |
| 157           | Off        | On         | On         | Off        | On         | Off        |
| 163           | Off        | On         | On         | Off        | On         | On         |
| 169           | Off        | On         | On         | On         | Off        | Off        |
| 175           | Off        | On         | On         | On         | Off        | On         |
| 181           | Off        | On         | On         | On         | On         | Off        |
| 187           | Off        | On         | On         | On         | On         | On         |
| 193           | On         | Off        | Off        | Off        | Off        | Off        |
| 199           | On         | Off        | Off        | Off        | Off        | On         |
| 205           | On         | Off        | Off        | Off        | On         | Off        |
| 211           | On         | Off        | Off        | Off        | On         | On         |
| 217           | On         | Off        | Off        | On         | Off        | Off        |
| 223           | On         | Off        | Off        | On         | Off        | On         |
| 229           | On         | Off        | Off        | On         | On         | Off        |
| 235           | On         | Off        | Off        | On         | On         | On         |
| 241           | On         | Off        | On         | Off        | Off        | Off        |
| 247           | On         | Off        | On         | Off        | Off        | On         |
| 253           | On         | Off        | On         | Off        | On         | Off        |
| 259           | On         | Off        | On         | Off        | On         | On         |
| 269           | On         | Off        | On         | On         | Off        | Off        |
| 271           | On         | Off        | On         | On         | Off        | On         |
| 277           | On         | Off        | On         | On         | On         | Off        |
| 283           | On         | Off        | On         | On         | On         | On         |
| 289           | On         | On         | Off        | Off        | Off        | Off        |
| 295           | On         | On         | Off        | Off        | Off        | On         |
| 301           | On         | On         | Off        | Off        | On         | Off        |
| 307           | On         | On         | Off        | Off        | On         | On         |
| 313           | On         | On         | Off        | On         | Off        | Off        |
| 319           | On         | On         | Off        | On         | Off        | On         |
| 325           | On         | On         | Off        | On         | On         | Off        |
| 331           | On         | On         | Off        | On         | On         | On         |
| 337           | On         | On         | On         | Off        | Off        | Off        |
| 343           | On         | On         | On         | Off        | Off        | On         |
| 349           | On         | On         | On         | Off        | On         | Off        |
| 355           | On         | On         | On         | Off        | On         | On         |
| 361           | On         | On         | On         | On         | Off        | Off        |
| 367           | On         | On         | On         | On         | Off        | On         |
| 373           | On         | On         | On         | On         | On         | Off        |
| 379           | On         | On         | On         | On         | On         | On         |

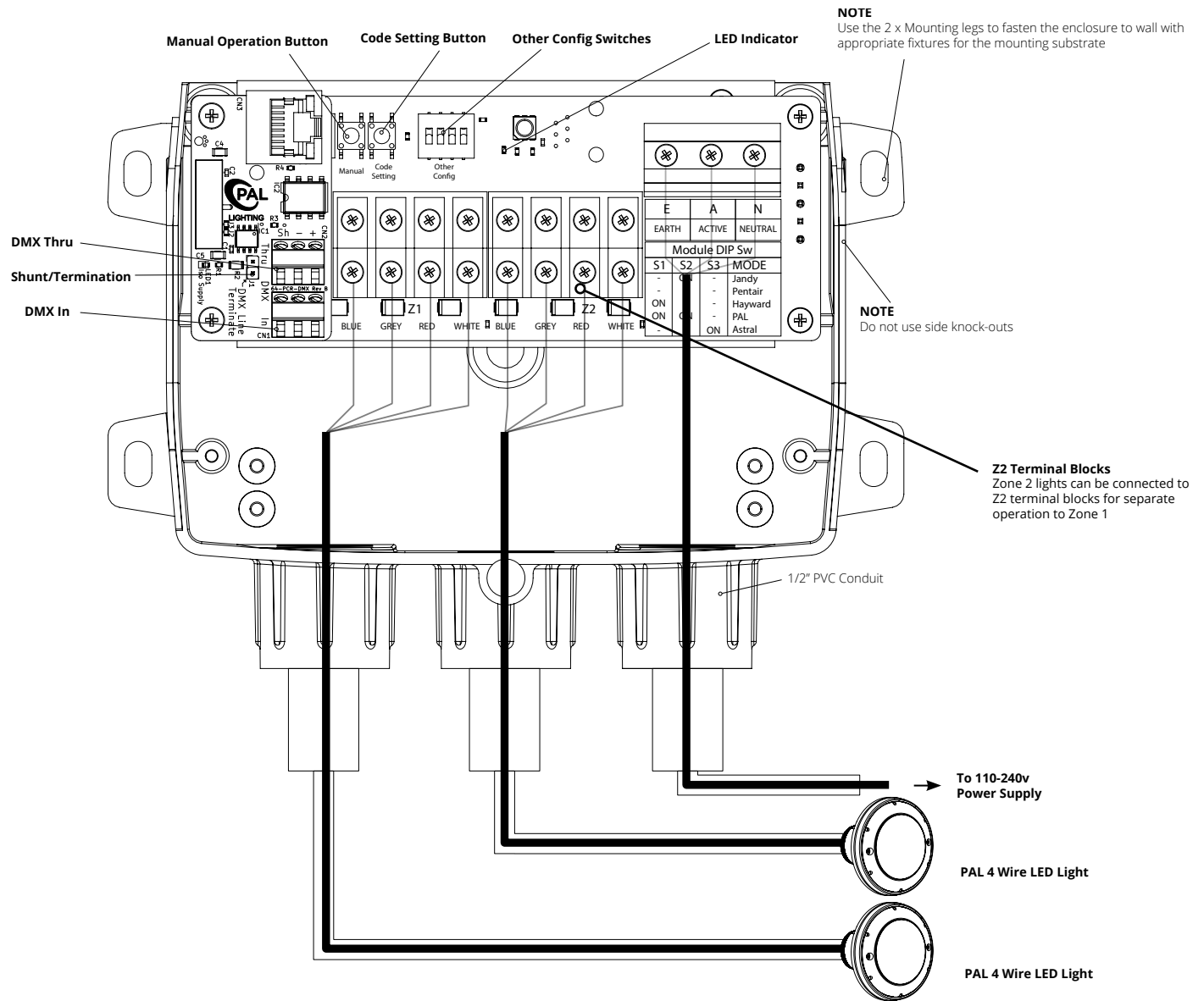


### V3 PCR-2DMX DRIVER/RECEIVER - 4 Channel addresses (Dipswitch #7 is off)

| Start Address | SW3 #1<br>"1" | SW3 #2<br>"2" | SW3 #3<br>"3" | SW2 #1<br>"4" | SW2 #2<br>"5" | SW2 #3<br>"6" |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 1             | Off           | Off           | Off           | Off           | Off           | Off           |
| 9             | Off           | Off           | Off           | Off           | Off           | On            |
| 17            | Off           | Off           | Off           | Off           | On            | Off           |
| 25            | Off           | Off           | Off           | Off           | On            | On            |
| 33            | Off           | Off           | Off           | On            | Off           | Off           |
| 41            | Off           | Off           | Off           | On            | Off           | On            |
| 49            | Off           | Off           | Off           | On            | On            | Off           |
| 57            | Off           | Off           | Off           | On            | On            | On            |
| 65            | Off           | Off           | On            | Off           | Off           | Off           |
| 73            | Off           | Off           | On            | Off           | Off           | On            |
| 81            | Off           | Off           | On            | Off           | On            | Off           |
| 89            | Off           | Off           | On            | Off           | On            | On            |
| 97            | Off           | Off           | On            | On            | Off           | Off           |
| 105           | Off           | Off           | On            | On            | Off           | On            |
| 113           | Off           | Off           | On            | On            | On            | Off           |
| 121           | Off           | Off           | On            | On            | On            | On            |
| 129           | Off           | On            | Off           | Off           | Off           | Off           |
| 137           | Off           | On            | Off           | Off           | Off           | On            |
| 145           | Off           | On            | Off           | Off           | On            | Off           |
| 153           | Off           | On            | Off           | Off           | On            | On            |
| 161           | Off           | On            | Off           | On            | Off           | Off           |
| 169           | Off           | On            | Off           | On            | Off           | On            |
| 177           | Off           | On            | Off           | On            | On            | Off           |
| 185           | Off           | On            | Off           | On            | On            | On            |
| 193           | Off           | On            | On            | Off           | Off           | Off           |
| 201           | Off           | On            | On            | Off           | Off           | On            |
| 209           | Off           | On            | On            | Off           | On            | Off           |
| 217           | Off           | On            | On            | Off           | On            | On            |
| 225           | Off           | On            | On            | On            | Off           | Off           |
| 233           | Off           | On            | On            | On            | Off           | On            |
| 241           | Off           | On            | On            | On            | On            | Off           |
| 249           | Off           | On            | On            | On            | On            | On            |
| 257           | On            | Off           | Off           | Off           | Off           | Off           |
| 265           | On            | Off           | Off           | Off           | Off           | On            |
| 273           | On            | Off           | Off           | Off           | On            | Off           |
| 281           | On            | Off           | Off           | Off           | On            | On            |
| 289           | On            | Off           | Off           | On            | Off           | Off           |
| 297           | On            | Off           | Off           | On            | Off           | On            |
| 305           | On            | Off           | Off           | On            | On            | Off           |
| 313           | On            | Off           | Off           | On            | On            | On            |
| 321           | On            | Off           | On            | Off           | Off           | Off           |
| 329           | On            | Off           | On            | Off           | Off           | On            |
| 337           | On            | Off           | On            | Off           | On            | Off           |
| 345           | On            | Off           | On            | Off           | On            | On            |
| 353           | On            | Off           | On            | On            | Off           | Off           |
| 361           | On            | Off           | On            | On            | Off           | On            |
| 369           | On            | Off           | On            | On            | On            | Off           |
| 377           | On            | Off           | On            | On            | On            | On            |
| 385           | On            | On            | Off           | Off           | Off           | Off           |
| 393           | On            | On            | Off           | Off           | Off           | On            |
| 401           | On            | On            | Off           | Off           | On            | Off           |
| 409           | On            | On            | Off           | Off           | On            | On            |
| 417           | On            | On            | Off           | On            | Off           | Off           |
| 425           | On            | On            | Off           | On            | Off           | On            |
| 433           | On            | On            | Off           | On            | On            | Off           |
| 441           | On            | On            | Off           | On            | On            | On            |
| 449           | On            | On            | On            | Off           | Off           | Off           |
| 457           | On            | On            | On            | Off           | Off           | On            |
| 465           | On            | On            | On            | Off           | On            | Off           |
| 473           | On            | On            | On            | Off           | On            | On            |
| 481           | On            | On            | On            | On            | Off           | Off           |
| 489           | On            | On            | On            | On            | Off           | On            |
| 497           | On            | On            | On            | On            | On            | Off           |
| 505           | On            | On            | On            | On            | On            | On            |



## V2 16/35/65 WATT PCR-2DMX DRIVER/RECEIVER



Note 1:  
The PCR-2DMX is suitable for PAL 4 wire lights.

Note 2:  
Knock outs are provided in the enclosure to connect up to 2 x PAL 4 wire UL Listed Pool Lights.

Note 3:  
An unlimited number of drivers can be controlled by one transmitter.

### PCR DMX Operation

The DMX input module allows remote DMX control over the two zone outputs in the PCR 2Z.

### Installation

1. Connect DMX input to “In” screw terminal block, green connections as indicated on the PCB.
2. Connect DMX loop to “Thru” screw terminal block.
3. If no loop out is used, place supplied shunt across “Term” pins.
4. Set DMX address as per the table on the following page.

### Operation

The DMX footprint of the PCR DMX is of two x 3 channel generic RGB lights, total 6 active channels. In 4 channel mode there is a vacant channel after each Blue, so each PCR4 occupies 8 channels. Refer to table below.

### Signal LED

When in DMX mode the “CPU” Led on the PCB operates as a DMX signal indicator. This LED is illuminated during the DMX packet – its flash rate is therefor dependent upon the DMX controller being used. At high rates of refresh the signal LED may not appear to flash.

### Loss of DMX

The output levels are held for 60 seconds after the DMX is lost, after which time the output is driven off. After a further 60 seconds of no DMX the fan (if fitted) is turned off.

### Other

The software includes temperature limiting – if the temperature goes over about 60C the output levels are reduced.

The PCR DMX input does not support RDM.

## V2 16/35/65 WATT PCR-2DMX DRIVER/RECEIVER DIP SWITCH ADDRESS SETTING

| 2Zone Driver # | 2Zone Start Address | "Module Config" Dip Switch #1 | "Module Config" Dip Switch #2 | "Module Config" Dip Switch #3 | "Module Config" Dip Switch #4 | "Other Config" Dip Switch #1 | "Other Config" Dip Switch #2 | "Other Config" Dip Switch #3 | "Other Config" Dip Switch #4 |
|----------------|---------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
| 1              | 1                   | On                            | On                            | Off                           | Off                           | Off                          | Off                          | Off                          | Off                          |
| 2              | 7                   | On                            | On                            | Off                           | Off                           | Off                          | Off                          | Off                          | On                           |
| 3              | 13                  | On                            | On                            | Off                           | Off                           | Off                          | Off                          | On                           | Off                          |
| 4              | 19                  | On                            | On                            | Off                           | Off                           | Off                          | Off                          | On                           | On                           |
| 5              | 25                  | On                            | On                            | Off                           | Off                           | Off                          | On                           | Off                          | Off                          |
| 6              | 31                  | On                            | On                            | Off                           | Off                           | Off                          | On                           | Off                          | On                           |
| 7              | 37                  | On                            | On                            | Off                           | Off                           | Off                          | On                           | On                           | Off                          |
| 8              | 43                  | On                            | On                            | Off                           | Off                           | Off                          | On                           | On                           | On                           |
| 9              | 49                  | On                            | On                            | Off                           | Off                           | Off                          | Off                          | Off                          | Off                          |
| 10             | 55                  | On                            | On                            | Off                           | Off                           | On                           | Off                          | Off                          | On                           |
| 11             | 61                  | On                            | On                            | Off                           | Off                           | On                           | Off                          | On                           | Off                          |
| 12             | 67                  | On                            | On                            | Off                           | Off                           | On                           | Off                          | On                           | On                           |
| 13             | 73                  | On                            | On                            | Off                           | Off                           | On                           | On                           | Off                          | Off                          |
| 14             | 79                  | On                            | On                            | Off                           | Off                           | On                           | On                           | Off                          | On                           |
| 15             | 85                  | On                            | On                            | Off                           | Off                           | On                           | On                           | On                           | Off                          |
| 16             | 91                  | On                            | On                            | Off                           | Off                           | On                           | On                           | On                           | On                           |
| 17             | 97                  | On                            | On                            | Off                           | On                            | Off                          | Off                          | Off                          | Off                          |
| 18             | 103                 | On                            | On                            | Off                           | On                            | Off                          | Off                          | Off                          | On                           |
| 19             | 109                 | On                            | On                            | Off                           | On                            | Off                          | Off                          | On                           | Off                          |
| 20             | 115                 | On                            | On                            | Off                           | On                            | Off                          | Off                          | Off                          | On                           |
| 21             | 121                 | On                            | On                            | Off                           | On                            | Off                          | On                           | Off                          | Off                          |
| 22             | 127                 | On                            | On                            | Off                           | On                            | Off                          | On                           | On                           | On                           |
| 23             | 133                 | On                            | On                            | Off                           | On                            | Off                          | On                           | On                           | Off                          |
| 24             | 139                 | On                            | On                            | Off                           | On                            | Off                          | On                           | Off                          | On                           |
| 25             | 145                 | On                            | On                            | Off                           | On                            | Off                          | Off                          | Off                          | Off                          |

# TERMS AND CONDITIONS

## PAL PRODUCT WARRANTY DETAILS

The PAL product is guaranteed against defects in materials and/or workmanship, under normal domestic use for (3) years from date of purchase.

Except as otherwise provided for by law, this guarantee becomes invalid if in the opinion of the Company the product has been misused, abused, damaged in transit, incorrectly installed, or connected to an electrical supply not corresponding with the products specification or a non-approved PAL power supply.

The Company is not responsible for forwarding freight charges or losses in transit.

If any part of the product proves to be defective within the warranty period, the defective part will be repaired or replaced, free of charge and at the discretion of PAL Lighting.

PAL Lighting shall not be liable for any loss of profits or any consequential, indirect or special loss, damage or injury of any kind whatsoever arising directly or indirectly from PAL Lighting products.

PLEASE REGISTER YOUR PRODUCT WARRANTY DETAILS ONLINE AT:

<https://pallighting.com/product-registration>

OR You MUST retain some proof of purchase to provide evidence of the DATE of the original purchase.